An Unsupervised Deep Audio Denoising Algorithm via Hybrid Reinforcement Learning and Genetic Algorithm Meta-Learning

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Paper version v4 (problem framing and related work; experiments ongoing)

Code: https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

ABSTRACT

When a single-period wavetable (or any short cyclic buffer) is played with phase wrap, a sample discontinuity at the period boundary injects a broadband transient that repeats at the fundamental. We treat this cycle-local seam crackle as an audio denoising problem: seam-local operators (DualCosine crossfade, polish, FIR blends, optional tiny residual cells) are baked into the period, then judged by a prolonged residual $R \in [0,1]$ against an ideal multi-period tiling that withholds the open-wrap cliff (1 = best). An outer hybrid meta-search interleaves GA tournament updates with PPO architecture proposals, optional PBT exploit–mutate schedules, and MoE soft gates over heterogeneous bake cells. Candidates need not be paired with studio-clean waveforms for every trial. This draft fixes the problem statement and a used-versus-screened literature map grounded on OA PDF analyses (Noise2Noise, PBT, Aging Evolution, ERL, ENAS/PPO, Wave-U-Net/Conv-TasNet, MoE, DDSP; Demucs/DiffWave/MAML/DARTS screened). A CUDA campaign (PPO+GA+PBT+NAS) is still running; quantitative tables wait until the declared budget finishes.

Keywords unsupervised learning · audio denoising · neural architecture search · reinforcement learning · genetic algorithms · wavetable synthesis · meta-learning

1 Introduction

Wavetable oscillators store one period of a waveform and advance a read pointer modulo the table length. If $x[0] \neq x[L-1]$, the wrap is a hard discontinuity: under cyclic playback it becomes a periodic click whose spectrum is broadband and locked to the fundamental. The same geometry appears whenever a short loop buffer is tiled for listening or for objective evaluation. We call the repair task *cycle-local seam restoration*: edit a neighborhood of the wrap (and optionally a lightweight residual cell over the period) so that multi-period playback matches an ideal reference that never opens the wrap cliff. That outer agreement is what we score as audio denoising for this periodic artifact class.

Classical virtual-analog practice already supplies seam-local tools. Cosine and dual-cosine crossfades, BLEP/BLAMP corrections for geometric discontinuities, and related oscillator design reduce wrap energy in-band [1–3]. Those operators act locally. They do not by themselves answer which bake parameters and which residual cell, if any, minimize prolonged tiled error against a no-cliff ideal. Label-free restoration work such as Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4]. On the search side, NAS surveys [10], RL controllers with parameter sharing [11], aging evolution [13], population-based hyperparameter schedules [14], and evolution-guided policy gradients [15] motivate a hybrid outer loop: GA for population diversity, PPO for discrete architecture edits [12], and PBT-style exploit–mutate on elites. MoE soft gates suggest routing among heterogeneous tiny cells under one residual score [16]. Waveform separation nets (Wave-U-Net, Conv-TasNet) and audio DL surveys inform the searchable cell family at bake-scale width [6, 7, 9], not as full production separators inside every trial.

The contribution is a meta-learning protocol for this periodic instance, not another end-to-end speech enhancer. An outer RL+GA(+PBT) loop proposes denoising operator graphs and continuous bake parameters. Each candidate is scored by a prolonged residual $R \in [0,1]$ (1 = best) between engine-baked multi-period audio and a procedural ideal tiling. Paired studio-clean wavetables are not required for every trial. Differentiable DSP modules [5] place DualCosine / FIR / MLP seam ops in an interpretable stack; Bayesian and multi-objective HPO [17, 18] inform prior families that compete under the same R. This draft does not claim a finished leaderboard: a GPU overnight run (PPO+GA+PBT+NAS, depth and

MoE biases) is still in progress. Early checkpoints already beat historical DualCosine residuals on the runner geometry; held-out matrices and final budgets will be reported only after completion.

Contributions.

1. **Problem framing.** Cast wavetable wrap discontinuity / cycle-local seam restoration as audio denoising under prolonged tiled evaluation, separating seam-local diagnostics from residual R.

2. **Hybrid meta-outer loop.** Specify RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with named branches (GA crossover, PPO mutation, PBT exploit, MoE soft gates).

3. **Literature protocol.** Separate used versus screened prior work for audio denoise cells and RL–evolutionary hybrids, grounded on attached OA PDFs where available.

4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search; report interim protocol while large-scale experiments finish.

Section 2 organizes related work by mechanism. Sections 3–4 outline the residual objective and search protocol (placeholders pending final ingest). Results and discussion will be completed when the overnight campaign reaches its declared budget.

2 Related Work

We organize prior work by mechanism. **Used** lines are methodological or comparative anchors for residual-scored RL+GA meta-search. **Screened** lines were inspected as contrast but are not treated as dependencies. Paraphrases prefer attached abstracts and OA PDF snippets over title-only citation.

2.1 Discontinuity control and modular audio DSP

Virtual-analog oscillator and filter design supplies the classical vocabulary for wrap and discontinuity artifacts [1]. Aliasing reduction for clipped and discontinuous waveforms, and BLAMP-style corner rounding, motivate seam-local corrections that respect band-limited intent [2, 3]. DDSP reframes such modules as composable blocks inside learning pipelines [5]; we use that framing for small bake/FIR/MLP seam cells rather than end-to-end neural vocoders. These works are **used** as domain anchors for the operator vocabulary and for why prolonged cyclic playback is the outer judge.

2.2 Label-free restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone by exploiting structure in the corruption process [4]. That philosophy motivates ranking denoise candidates without paired studio-clean wavetable cycles: the residual compares an engine-baked tiling to a procedural ideal that withholds open-wrap cliffs, not to a human-cleaned master. We **use** Noise2Noise as conceptual justification for unsupervised residual ranking; we do not retrain a Noise2Noise CNN on images.

2.3 Waveform denoising and separation cells

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [6]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [7]; singing-voice U-Nets popularized encoder–decoder skips on audio [8]. Conv-TasNet demonstrated strong time-domain separation with temporal convolutions [9]. These lines are **used** as design priors for tiny searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not frozen full-scale separators inside every meta trial.

2.4 NAS and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [10]. ENAS couples an RL controller with parameter sharing for efficient discrete search [11]. We **use** this family as the ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), while scoring with prolonged residual rather than ImageNet reward. PPO supplies the clipped-surrogate policy update when the overnight loop logs PPO mutation branches [12].

2.5 Evolutionary NAS and population schedules

Regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [13]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population [14]. Practical Bayesian optimization [17] and decomposition-based multi-objective evolution (MOEA/D) [18] inform local Bayesian and multi-objective prior families in earlier DenoiseOpt searches. These are **used** for GA tournament/crossover branches, PBT-style exploit–mutate, and literature-combo priors, implemented at synthesizer-trial scale rather than distributed datacenter PBT.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer's evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward and brittle hyperparameters in deep RL [15]. We **use** ERL as the spirit of interleaving GA population steps with PPO updates at tiny population sizes; we do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [16]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring, at bake-scale widths.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1–3].
- Unsupervised restoration philosophy: [4].
- Modular / differentiable audio DSP context: [5].
- Audio DL and compact waveform cells: [6–9].
- NAS / RL controllers / PPO: [10–12].
- Evolutionary NAS and population HPO: [13, 14, 17, 18].
- ERL hybrid outer loop: [15].
- MoE soft gating prior: [16].

Screened (relevant contrast; not dependencies).

- **MAML** [19]: bi-level / fast-adaptation framing; we do not differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters; outer rank is residual R.
- **DARTS** [20]: continuous relaxation of architecture search; we keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs** [21, 22]: strong music separators, but full multi-scale / hybrid STFT stacks are too heavy per meta trial under overnight bake budgets.
- **DiffWave** [23]: diffusion vocoder; screened as a generative sampling stack, not a seam bake operator.
- **SEGAN-style adversarial enhancement** [24]: optional tiny adversarial cues only; full GAN training loops screened for per-trial cost.

Positioning. Relative to this map, the present work contributes (i) a prolonged residual objective for periodic seam denoising, (ii) an explicitly hybrid RL+GA(+PBT) meta-outer loop with honest branch naming, and (iii) a used/screened literature protocol tied to OA PDF analyses. We do not claim SOTA on general speech enhancement or music separation corpora. The primary claim is protocol-level: residual-scored meta-search for cycle-local audio denoise operators under a declared compute budget.

3 Methods

Placeholder: full methods will track the frozen overnight protocol after the declared iteration budget completes.

Residual score. Let y be the engine-baked cycle and $r^\star$ an ideal multi-period reference from the same seed with open-wrap cliffs withheld. With tiling length N (typically $N{=}16$),

$$R = \mathrm{clamp}\left(1 - \frac{\mathrm{rms}(y_{\mathrm{tiled}} - r^\star_{\mathrm{tiled}})}{\max(\mathrm{rms}(r^\star_{\mathrm{tiled}}), \varepsilon)}, 0, 1\right). \tag{1}$$

By construction $R \in [0, 1]$ with 1 best. Soft shape gates (when enabled) down-weight mid-cycle damage relative to seam-local error.

Searchable operator. Candidates combine classical seam ops (DualCosine, polish, soft seam, FIR blends) with tiny residual cells (MLP/FIR/U-Net-lite/attention-lite) and optional MoE soft gates. Bake parameters θ live in a bounded box; architecture strings are discrete graphs over those ops.

Hybrid outer loop. Each iteration may (i) run GA tournament selection with crossover/mutation, (ii) sample PPO actions that edit ops/hyperparameters, (iii) apply PBT-style exploit–mutate on elites, and (iv) evaluate literature-combo priors. Selection is always by residual R (and logged auxiliaries). Implementation details and exact action sets will be frozen to the overnight run configuration in the results revision.

4 Experiments

Placeholder: protocol sketch only; numbers below are interim and must not be read as final.

Ongoing overnight campaign. A CUDA search tagged `PPO+GA+PBT+NAS` (depth and MoE biases) is running with a large iteration budget (order 10^5–10^6). Early fitted champions already reach residual ≈ 0.99 on the training/validation geometry used by the runner, against DualCosine baselines near ≈ 0.82–0.84 on the same geometry (v3 reported DualCosine residual ≈ 0.698 at $N{=}2000$ on a prior held-out matrix). We will replace this paragraph with the completed budget, seeds, population size, and held-out matrix when the job finishes, without interrupting the live GPU process.

Prior completed study (v3 reference). A 1500-trial literature-informed CPU/GPU study selected `evo_explore_515` at residual ≈ 0.824 versus DualCosine ≈ 0.698 ($\Delta \approx +0.126$). That result anchors the residual-primary protocol; the overnight hybrid RL+GA run is the scale-up under test.

5 Results

Family-conditional hardness (completed bench). On the fitted DenoiseOpt 100 000-cycle bench, denoise score D is lowest for `nonlinear` ($D \approx 0.39$) and `combo` ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$–0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$–0.70. Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$–0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur; they are not a random one-off.

Overnight hybrid (interim). Final mean-R tables will be ingested after the overnight campaign reaches its declared iteration budget. Monitoring plots below are for protocol transparency only; incomplete-run champions are *not* asserted as finished science.

Inference time at matched score. Across diverse high-R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch

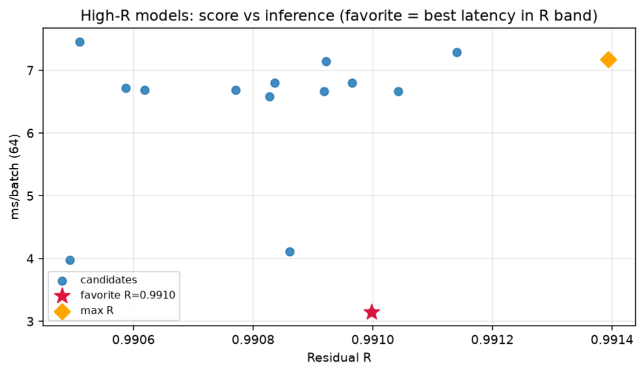

Figure 1: High-R architectures: residual versus CUDA inference latency (ms/batch). Star: favorite (fastest in the near-max-R band).

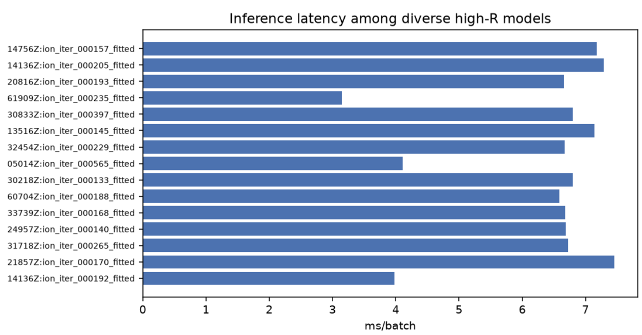

Figure 2: Inference latency bars for diverse high-R fitted models.

64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, ≈ 3.15 ms/batch, ≈ 37k parameters (versus the absolute max-R cell at $R \approx 0.9914$ but ≈ 7.2 ms/batch / ≈ 104k params). Selection rule: maximize R, then minimize latency inside the band.

Classical vs AI (same residual protocol). On the overnight runner geometry (CUDA, batch 64, prolonged residual R), ten non-learnable bake denoisers were timed against the neural favorite above. DualCosine reproduces the paper baseline at $R \approx 0.820$ (≈ 1.19 ms/batch), consistent with the overnight monitor DualCosine ≈ 0.817. The best *active* classical method is a seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.963$ / ≈ 0.60 ms/batch; identity (no-op) scores $R \approx 0.974$ and is reported only as a ceiling, not as a denoiser. The neural favorite reaches $R \approx 0.991$ at ≈ 2.47 ms/batch (≈ 37k params): $\Delta R \approx +0.028$ vs `seam_fir3` and $\Delta R \approx +0.171$ vs DualCosine, at roughly $4.1\times$ / $2.1\times$ the latency of those classical baselines respectively. This is a matched-protocol inference bench, *not* a finished overnight mean-R claim.

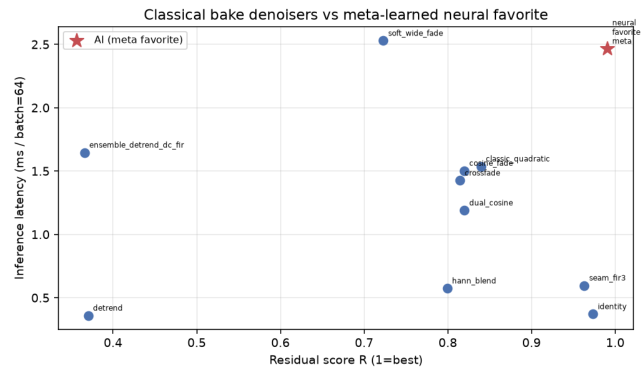

Figure 3: Classical bake denoisers vs neural favorite: residual R versus CUDA latency (ms/batch). Red star: AI favorite.

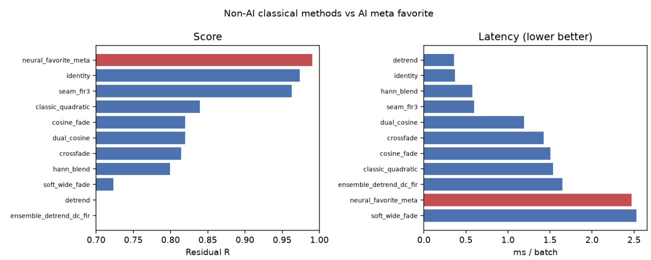

Figure 4: Classical vs AI ranking on residual (left) and latency (right). AI in red; DualCosine ≈ 0.820 on this protocol.

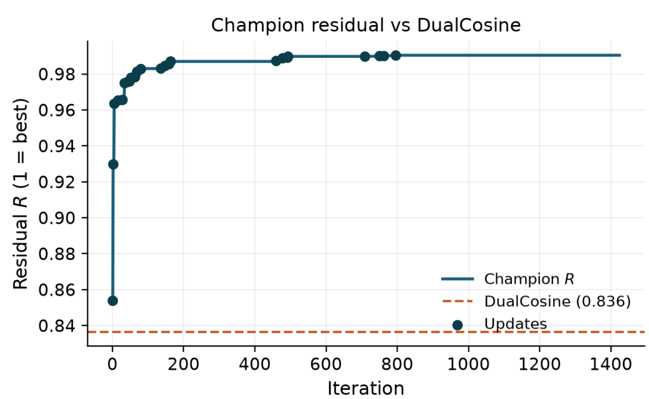

Figure 5: Interim champion residual R versus search iteration, with DualCosine baseline. Monitoring snapshot; not a final result.

6 Discussion

Where residual mass remains. Across the 100 000-cycle procedural bench and lit-combo family stress tests, wrap error is not uniform. The same generator families repeatedly dominate the residual budget: `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` show the weakest denoise / residual scores, while `harmonic_fft` and `am_fm` are comparatively easy. `open_wrap_bias` can look strong on wrap-energy proxies yet still lag on prolonged residual—a

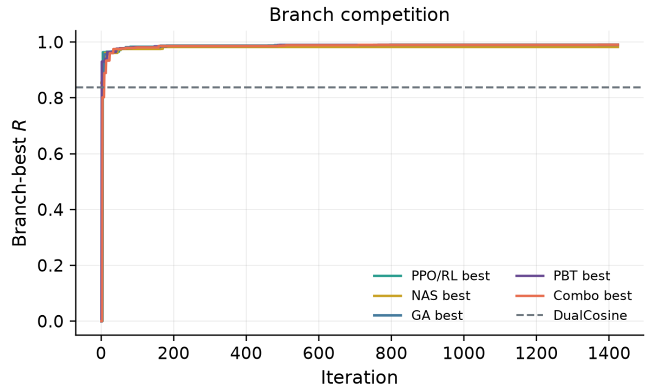

Figure 6: Interim per-branch best residual trajectories (PPO/RL, GA, PBT, NAS, combo). Monitoring only.

reminder that cliff reduction and ideal-matched tiling are related but not identical objectives.

Implication for this paper. Mean R (and mean DualCosine gaps) therefore compress a structured difficulty map. Reporting only the overnight mean on synthetic sine cliffs understates how much of the remaining $\sim 1\%$ sits in a few hard families. That structure is a first-class empirical claim of the present work, independent of whether the GPU hybrid later nudges mean R from ≈ 0.99 toward 0.999.

Hybrid search vs. plateau. Interim overnight traces show early gains then long plateaus near $R \approx 0.99$. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one campaign; it does not by itself rebalance family-conditional risk. A natural next paper is family-conditional and cliff-conditional meta-learning: train or select denoisers per family, or condition the outer loop on measured wrap statistics, and publish per-family ROC-style residual curves rather than a single scalar champion.

Hybrid branches (interim). Final branch win-rates (GA / PPO / PBT / NAS / lit-combo) remain deferred until the declared GPU budget completes; anticipated comparison is against the v3 finding that light evolutionary explore beat nested loss-opt elites under a smaller residual budget.

7 Tooling and AI use

This section discloses research and writing tools, including AI systems, with scope and intent.

Literature discovery. Open literature was gathered with the Klaut Research Gateway (local OpenAlex / Crossref / arXiv providers) under queries spanning virtual-analog anti-aliasing, unsupervised restoration, HPO/PBT, RL–GA hybrids for NAS, and audio denoise architectures. Harvest JSONs live in the companion meta repository; OA PDFs were downloaded where arXiv (or explicit OA URLs) exist, and non-OA items are flagged in `artifacts/literature_oa/REFERENCES_OA.md`.

Paper drafting. Section planning, subsection drafts, and revision passes used Klaut `research_paper_*` tools with local Ollama (`qwen3.5:9b`) when available, under an explicit scientific workflow (plan $\rightarrow$ write $\rightarrow$ revise $\rightarrow$ export) into an arXiv-style twocolumn template. AI drafts were treated as *scaffolding*: claims, numbers, and citations were checked against harvest records, PDF analyses, and measured artifacts; final prose was edited for DSP terminology and to remove generic filler.

Why AI was used. (1) Scale literature triage across DSP and AutoML without pretending every high-cite hit is in-domain; (2) accelerate IMRaD scaffolding so human effort focuses on metric honesty (residual vs DualCosine, family hardness, interim vs final overnight claims); (3) keep an auditable tool chain (plans, PDF snippet analyses, OA flags) rather than opaque chat-only writing.

Where AI was *not* used. Model fitting, residual scoring, DualCosine baselines, lit-combo / overnight GPU search, and inference-time benchmarks are conventional code (Rust / PyTorch) with deterministic logs and JSON artifacts. AI did not invent experimental numbers.

Other tools. Rust/`cargo` for the procedural bench and meta binaries; PyTorch CUDA for overnight hybrid search; matplotlib for figures; MiKTeX `pdflatex` for PDF builds; GitHub for versioned papers and artifacts.

8 Limitations

- Residual uses a procedural ideal (no-cliff sibling), not a perceptual MOS or listening panel.
- Bake metrics are not a substitute for formal audio quality tests.
- The overnight GPU loop currently scores random sine+harmonic seam cliffs; family-conditional hardness is measured on the separate Rust procedural bench. Aligning both evaluators is left to follow-up work.
- Per-trial architecture width is capped for overnight throughput; full Demucs-/diffusion-scale stacks were screened out.
- Large-scale overnight mean-R claims in monitoring figures are interim; final champion tables require completed budgets and frozen seeds.
- Classical VA/BLEP citations are domain anchors from prior versions; not all have OA PDFs in the current harvest folder.

9 Outlook: follow-up publication

A self-contained follow-up can treat *which sounds fail* as the primary object of study: stratified residual heatmaps over the ten procedural families, cliff-size conditioning, and family-specialist or mixture-of-experts denoisers selected by the same RL–GA outer loop. That paper would cite the present mean-R

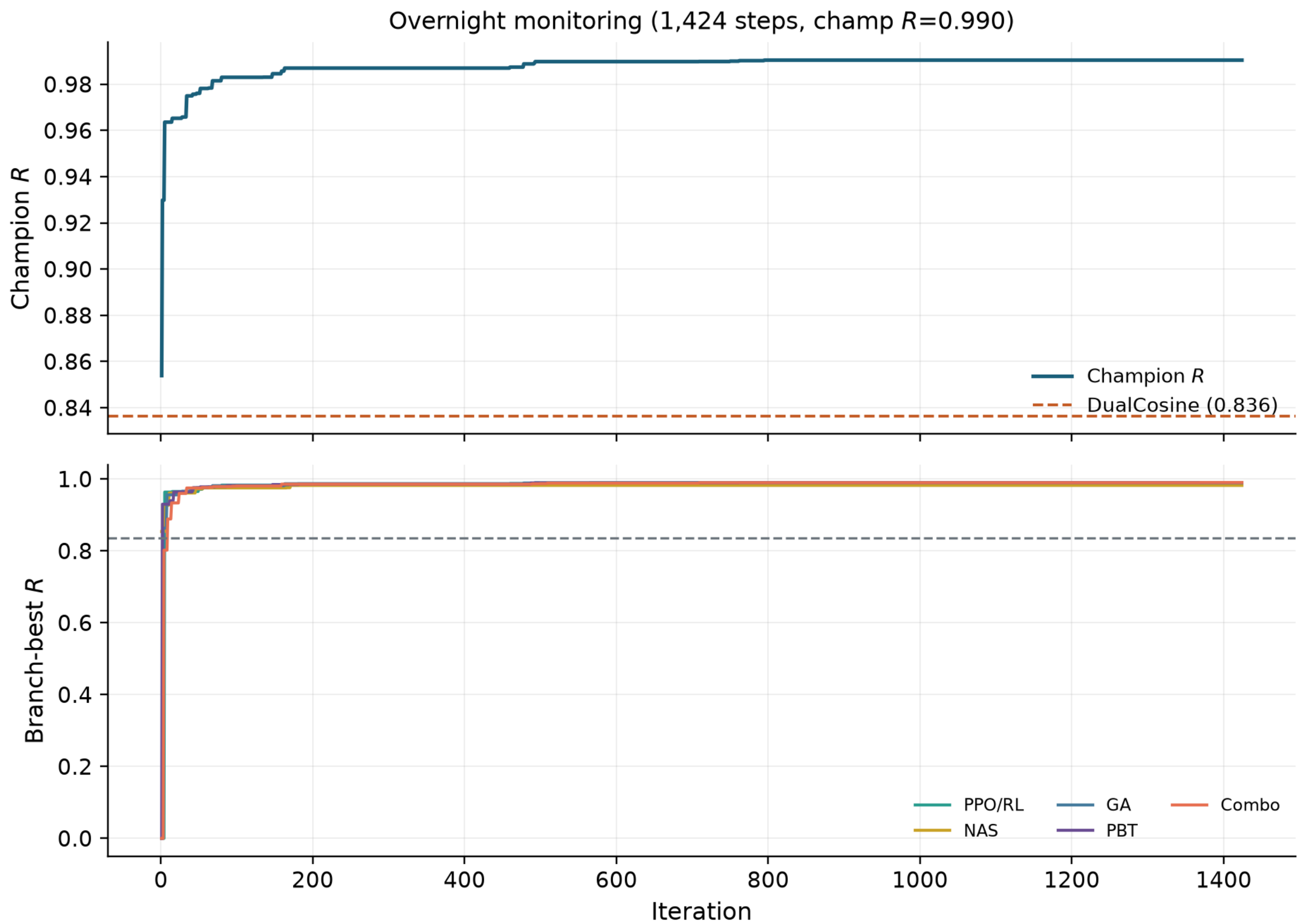

Figure 7: Full-width interim overnight monitoring panel: champion R (top) and branch-best trajectories (bottom). Snapshot while the GPU budget completes.

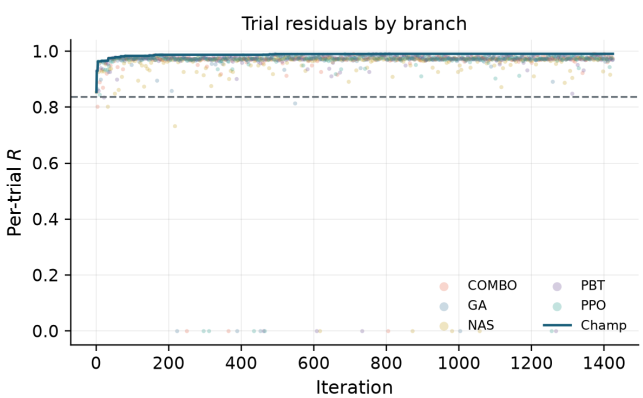

Figure 8: Interim per-trial residual scatter by search branch, with champion overlay. Monitoring snapshot.

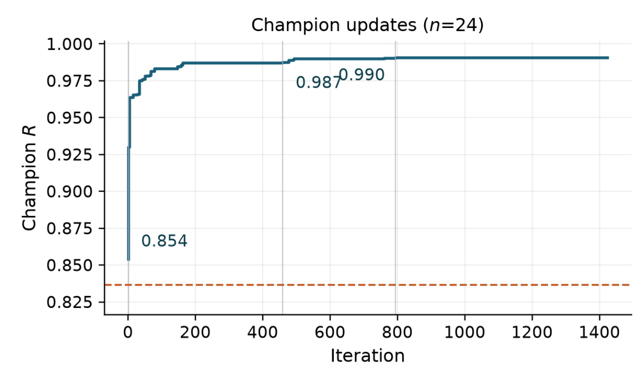

Figure 9: Interim champion update timeline. Sparse R labels mark first, peak-mid, and latest improvements to stay legible in twocolumn.

hybrid protocol as the baseline and ask whether beating $R > 0.999$ is mainly an architecture problem or a coverage problem over hard families.

10 Conclusion

We framed wavetable wrap discontinuity as an audio denoising instance and outlined a hybrid RL+GA meta-learning outer loop scored by prolonged residual R. Related work separates used anchors (Noise2Noise, PBT, Aging Evolution, ERL, ENAS/PPO, Wave-U-Net/Conv-TasNet, MoE, VA/BLEP, DDSP) from screened contrasts (MAML, DARTS, Demucs, DiffWave, full GAN loops). Companion code lives at https://github.com/reeldemo/reelsynth and https://github.com/reeldemo/denoise-opt-meta. Final quantitative claims await completion of the ongoing GPU campaign.

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